IMAGE PROCESSING TECHNIQUES – EXTENDED CLASS ACTIVITIES WORKSHEET

UG Level | Duration: 10 Minutes per Activity

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Activity 12: Morphological Operations (10 Minutes)

Tasks:

1. Define erosion.

2. Define dilation.

3. Which operation removes small noise?

4. What is a structuring element?

Answer:
1- Erosion is a morphological operation that removes pixels from the boundaries of objects, making them smaller or thinner.

2- Dilation is a morphological operation that adds pixels to the boundaries of objects, making them larger or thicker.

3- Opening operation removes small unwanted noise from the foreground.

4- A structuring element is a small predefined shape or matrix used in morphological operations such as erosion and dilation.